DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY

University Examinations 2014/2015

THIRD YEAR SEMESTER TWO EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

ICS 2311: COMPUTER GRAPHICS

DATE: 12TH AUGUST 2014 **TIME: 11.00AM – 1.00PM**

INSTRUCTIONS: ANSWER QUESTION 1 AND ANY OTHER TWO.

QUESTION ONE (30 MARKS)

a) Define refresh buffer/frame buffer. [2 marks]
b) Highlight any four key differences between a Random scan display and a Raster scan display [4 marks]
c) Highlight any two advantages and any two disadvantages of Liquid Crystal Display (LCD). [4 marks]
d) Describe the term "viewing pipeline" as used in computer graphics [2 marks]
e) Define the term "Aliasing" [2 marks]
f) Consider three different raster systems with resolutions of 640 x 480, 1280 x 1024, and 2560 x 2048. What size is frame buffer (in bytes) for each of these systems to store 12 bits per pixel? [4 marks]

g) Using appropriate illustrations, describe the following transformation techniques
i) Scaling [2 marks]
ii) Shearing [2 marks]
h) Describe the Liang Barsky Clipping algorithm [3 marks]
i) Identify any four applications of computer graphics. [2 marks]
j) Differentiate parallel projection from perspective projection [3 marks]

QUESTION TWO (20 MARKS)

a) Suppose (x1,y1) and (x2,y2) defines a line segment. Consider the left boundary of the clipping region x = Wxmin.

i) Specify the visible side of the boundary [2 marks]

ii) Construct a test to find whether both end-points are on the invisible, or both on the visible side of the boundary, or whether one point is one side and the other is the opposite side. [3 marks]

iii) Find the intersection of the line segment with the boundary, if any [5 marks]

b) A trapezium is a four-sided shape, with top and bottom sides horizontal. For example, it would have coordinates (going round anti-clockwise from the bottom left corner): (x1,y1), (x2,y1), (x3,y2), (x4,y2). In this case x1 < x2, and y2 > y1.

i) Construct a line clipping algorithm for the trapezium as clipping region.

[5 marks]

ii) Construct a polygon clipping algorithm on the lines of Sutherland-Hodgman for such a clipping region. [5 marks]

QUESTION THREE (20 MARKS)

a) Rotate point A (4, 5) through an angle of 30^0 at an arbitrary point B (3, 4) [5marks]

b) Show that a rotation about the origin is

$x' = x \cos \theta - y \sin \theta$

$y' = x \sin \theta + y \cos \theta$ [5 marks]

c) Describe Mid-Point Circle algorithm [10 marks]

QUESTION FOUR (20 MARKS)

The following is a snap shot of two graphics functions code. Study them and answer the questions that follow.

```
Line 1    void initGL() {
Line 2     glClearColor(0.0f, 0.0f, 0.0f, 1.0f); }

Line 3    void display() {
Line 4    glClear(GL_COLOR_BUFFER_BIT);

Line 5    glBegin(GL_QUADS);
Line 6    glColor3f(0.2f, 0.2f, 0.2f);
Line 7    glVertex2f(-0.9f, -0.7f);
Line 8     glColor3f(1.0f, 1.0f, 1.0f);
Line 9     glVertex2f(-0.5f, -0.7f);
Line 10   glColor3f(0.2f, 0.2f, 0.2f);
Line 11   glVertex2f(-0.5f, -0.3f);
Line 12   glColor3f(1.0f, 1.0f, 1.0f);
Line 13   glVertex2f(-0.9f, -0.3f);
Line 14    glEnd();

Line 15   glBegin(GL_TRIANGLES);
```

```
Line 16    glColor3f(0.0f, 0.0f, 1.0f);
Line 17    glVertex2f(0.1f, -0.6f);
Line 18   glVertex2f(0.7f, -0.6f);
Line 19  glVertex2f(0.4f, -0.1f);
 Line 20  glEnd();

Line 21   glBegin(GL_POLYGON);
Line 22    glColor3f(1.0f, 1.0f, 0.0f);
Line 23    glVertex2f(0.4f, 0.2f);
Line 24    glVertex2f(0.6f, 0.2f);
Line 25    glVertex2f(0.7f, 0.4f);
Line 26    glVertex2f(0.6f, 0.6f);
Line 27    glVertex2f(0.4f, 0.6f);
Line 28    glVertex2f(0.3f, 0.4f);
Line 29   glEnd();
Line 30   glFlush();
}
```

a) Briefly comment on lines of code numbered 2, 4, 5, 6, 7 and 30. Use line numbers in your answers. [6 marks]
b) Describe in details the expected output. [5 marks]
c) Sketch the expected output. [4 marks]
d) (i) How will you include the following title "Vertex, Primitive & color" to the output? [2 marks]
(ii) How will you call the sub routine "display" in the main method? [2 marks]
(iii) Identify the function in the main method that executes the application [1 mark]

QUESTION FIVE (20 MARKS)

a) Discuss the following hidden surface removal techniques
i) Back face detection method [5 marks]
ii) Depth Buffer method [5 marks]
b) Differentiate the following terms as used in graphics animation
i) Key frames [2.5 marks]
ii) Morphing [2.5 marks]
c) Describe boundary fill algorithm [5 marks]